

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: Am-241			FORMS: Soluble		
PHYSICAL CHARACTERISTICS HALF-LIFE: 432.2 y DECAY EMISSIONS					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV. Ave)	%	E (keV)	%
60	36	11*	12	5486	85
14	15	37*	11	5443	13
18	9	42*	10	5388	2
		38*	10		
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY C – level (low hazard): Up to 0.2 µCi B – level (moderate hazard): > 0.2 µCi, < 10 µCi A – level (high hazard): > 10 µCi					
EXTERNAL RADIATION HAZARDS Gamma dose rate, point source at 1 ft, 1 mCi: 0.24 mrem/h Beta dose to skin, point source at 1 ft, 1 mCi: 0 mrem/h Contamination skin dose, uniform deposit of 1 µCi/cm ² : 72 mrem/h			INTERNAL RADIATION HAZARDS <i>Alpha particles are an internal hazard</i> Annual Limit on Intake: 0.8 µCi (Ingestion) 0.006 µCi (Inhalation) *Highly radiotoxic metal due to alphas* The values above indicate the activity taken into the body that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE).		
SHIELDING Gammas/X-rays: 0.106 mm of lead will reduce the gamma dose rate by 90%. Betas/electrons: Shielding is unnecessary for electrons emitted from Am-241. Bremsstrahlung may be created and require shielding.			DOSIMETRY AND BIOASSAY REQs Urine assays will be required after spill or contamination incidents involving more than 0.1 µCi.		
SPECIAL PROBLEMS AND PRECAUTIONS: 1. Am-241 is highly radiotoxic. Do not manipulate Am-241 sealed/electroplated sources without Health Physics review. 2. If not sealed or electroplated, work in a fume hood to capture any dusts. 3. Recommended survey probe: Thin-window NaI 4. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often. 5. Survey work areas before, during, and after work. Instrument and smear surveys are required. 6. Segregate waste to those with half-lives of greater than 120 days (excluding H3 and/or C14). Survey the waste disposal area to ensure exposure rates are less than 2 mR/hr at 1 foot. 7. Limit soluble waste to the sewer to less than 0.001 µCi/day per lab.					

References:

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- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
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